
Traffic Accident Severity Prediction

Using Machine Learning

Problem Id: 20

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Problem Description

Traffic accidents are a major issue affecting road safety and public health. The severity of accidents depends on several factors such as vehicle speed, weather conditions, road type, and vehicle category. Predicting accident severity can help authorities and emergency services respond more efficiently and reduce damage and casualties.

This project aims to develop a machine learning system that predicts the severity of traffic accidents based on various input features. The system will use historical accident data and apply preprocessing techniques to clean the dataset and remove irrelevant records. Feature selection methods will be used to identify the most important factors affecting accident severity, such as speed and weather conditions.

Machine learning algorithms such as Logistic Regression and Support Vector Machine (SVM) will be used to build the prediction model. The SVM model will use different kernels such as Linear, Polynomial, and RBF to capture simple and complex patterns in the data. The performance of the model will be evaluated using metrics like confusion matrix and recall score to measure prediction accuracy.

Dataset Description

The dataset used in this project contains road accident records with information related to drivers, vehicles, road conditions, weather conditions, collision types, and accident severity. The dataset was used to predict whether an accident would result in slight injury or severe injury.

The dataset consists of 12,316 records and multiple categorical and numerical attributes. Important features include driver age, driving experience, type of vehicle, road alignment, weather conditions, type of collision, number of vehicles involved, and number of casualties.

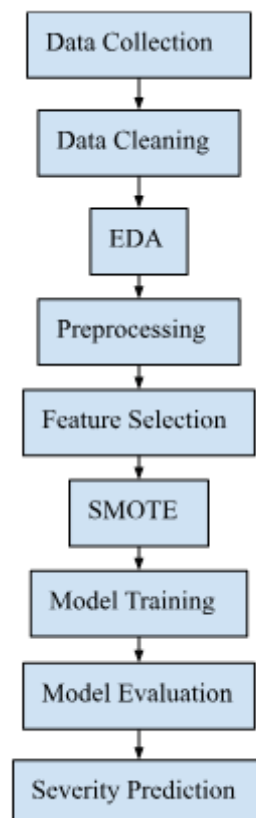
During preprocessing, missing values and inconsistent categories were handled, and category reduction techniques were applied to reduce feature complexity. The target variable used for classification was **Accident_severity**, which was grouped into two classes:

- Slight Injury
- Severe Injury

The dataset was highly imbalanced, with slight injury cases occurring more frequently than severe injury cases. Therefore, SMOTE (Synthetic Minority Oversampling Technique) was used to balance the dataset before model training.

The processed dataset was then used to train and evaluate different machine learning models including Support Vector Machine and Logistic Regression.

Flow Diagram



Modules for Project Implementation

Modules Used

1. pandas

Pandas was used for data manipulation, cleaning, preprocessing, and handling tabular datasets.

Functions used:

- `pd.read_csv()` – reads dataset from CSV file.
- `pd.DataFrame()` – creates dataframe objects.
- `df.drop()` – removes unnecessary columns.
- `df.fillna()` – handles missing values.
- `df.value_counts()` – counts category frequencies.
- `pd.get_dummies()` – performs one-hot encoding.

2. NumPy

NumPy was used for numerical operations and array handling.

Functions used:

- `np.array()` – creates arrays.
- `np.arange()` – generates numerical ranges.
- `np.random()` – generates random values for testing and visualization.

3. Matplotlib

Matplotlib was used for creating graphs and visualizations.

Functions used:

- `plt.figure()` – creates figure windows.
- `plt.bar()` – creates bar graphs.
- `plt.show()` – displays plots.
- `plt.title()` – sets graph titles.
- `plt.xlabel()` / `plt.ylabel()` – labels axes.

4. Seaborn

Seaborn was used for advanced statistical visualizations.

Functions used:

- `sns.countplot()` – visualizes categorical distributions.
- `sns.heatmap()` – creates correlation and confusion matrix heatmaps.
- `sns.boxplot()` – visualizes numerical distributions.
- `sns.violinplot()` – displays data density distributions.

5. scikit-learn

Scikit-learn was used for preprocessing, model training, evaluation, and feature selection.

Functions and Classes used:

- `train_test_split()` – splits dataset into training and testing sets.
- `StandardScaler()` – scales numerical features.
- `LabelEncoder()` – encodes categorical target labels.
- `SelectKBest()` – selects important features.
- `SVC()` – implements Support Vector Machine models.
- `LogisticRegression()` – implements Logistic Regression model.
- `accuracy_score()` – calculates accuracy.
- `classification_report()` – generates precision, recall, and F1-score.
- `confusion_matrix()` – generates confusion matrix.

6. imbalanced-learn

Imbalanced-learn was used to balance the dataset.

Functions used:

- SMOTE() – generates synthetic minority class samples.

7. joblib

Joblib was used to save and load trained models and preprocessing files.

Functions used:

- joblib.dump() – saves trained models and objects.
- joblib.load() – loads saved models and objects.

Platform Used:

Jupyter Notebook

Jupyter Notebook was used for the complete development and implementation of the project. It provides an interactive environment for writing, testing, and executing Python code in separate cells. Jupyter was especially useful for data preprocessing, exploratory data analysis (EDA), visualization, machine learning model training, and result analysis. Its notebook-based structure made experimentation, debugging, and visualization easier throughout the project development process.

Machine Learning Model Used

1. Support Vector Machine (SVM)

Support Vector Machine is a supervised machine learning classification algorithm used to classify accident severity into slight or severe injury categories. SVM works by finding the optimal decision boundary (hyperplane) that separates data points of different classes. Different kernel functions such as Linear, RBF, Polynomial, and Sigmoid were used to analyze model performance on nonlinear and complex accident data patterns.

2. Logistic Regression

Logistic Regression is a supervised classification algorithm used for binary classification problems. It predicts the probability of an accident belonging to a particular severity class using a logistic function. Logistic Regression was used as a baseline model to compare performance with SVM models because of its simplicity, interpretability, and efficient classification capability.

Evaluation metrics used

1. Accuracy

Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted observations to the total observations.

$$Accuracy = \frac{TN + TP}{TP + TN + FP + FN}$$

Where

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

2. Precision

Precision measures how many predicted positive cases were actually positive.

$$Precision = \frac{TP}{TP + FP}$$

Higher precision indicates fewer false positive predictions.

3. Recall

Recall measures how many actual positive cases were correctly identified by the model.

$$Recall = \frac{TP}{TP + FN}$$

Higher recall indicates better detection of severe accident cases.

4. F1-Score

F1-Score is the harmonic mean of Precision and Recall. It provides a balanced measure of model performance.

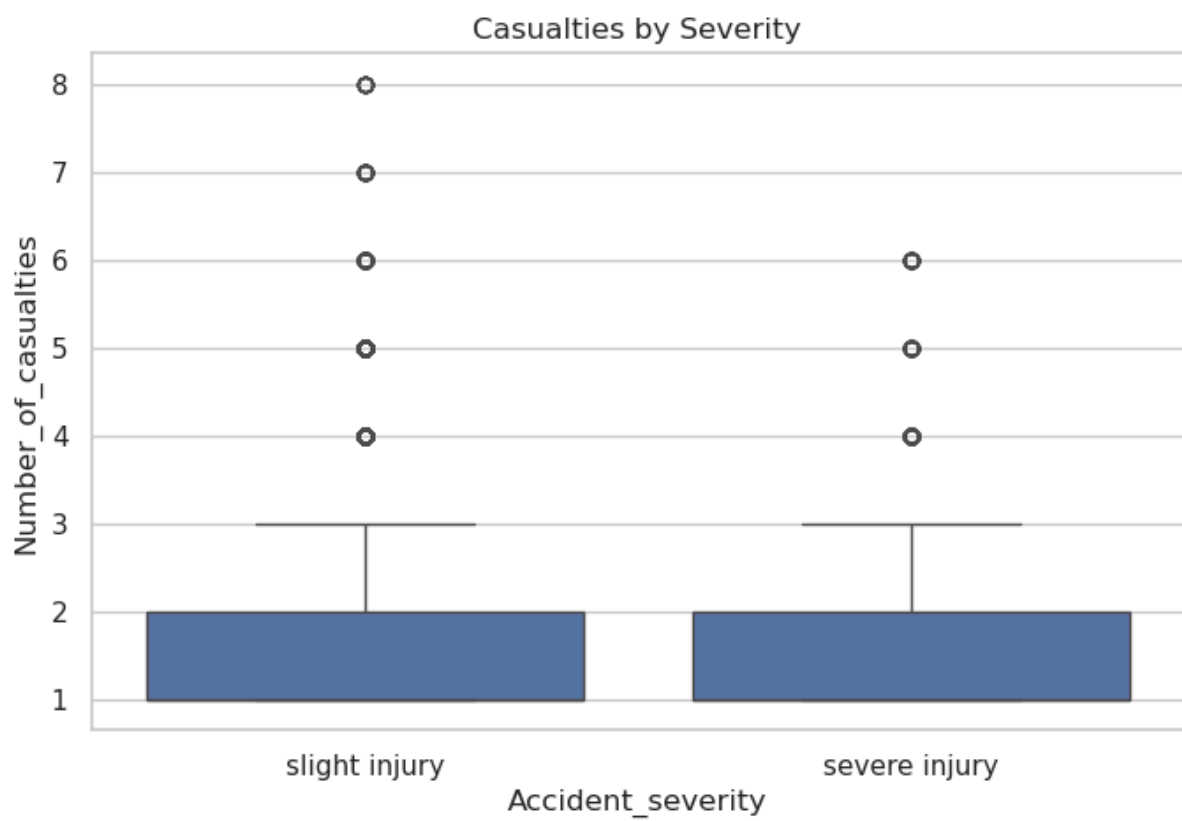
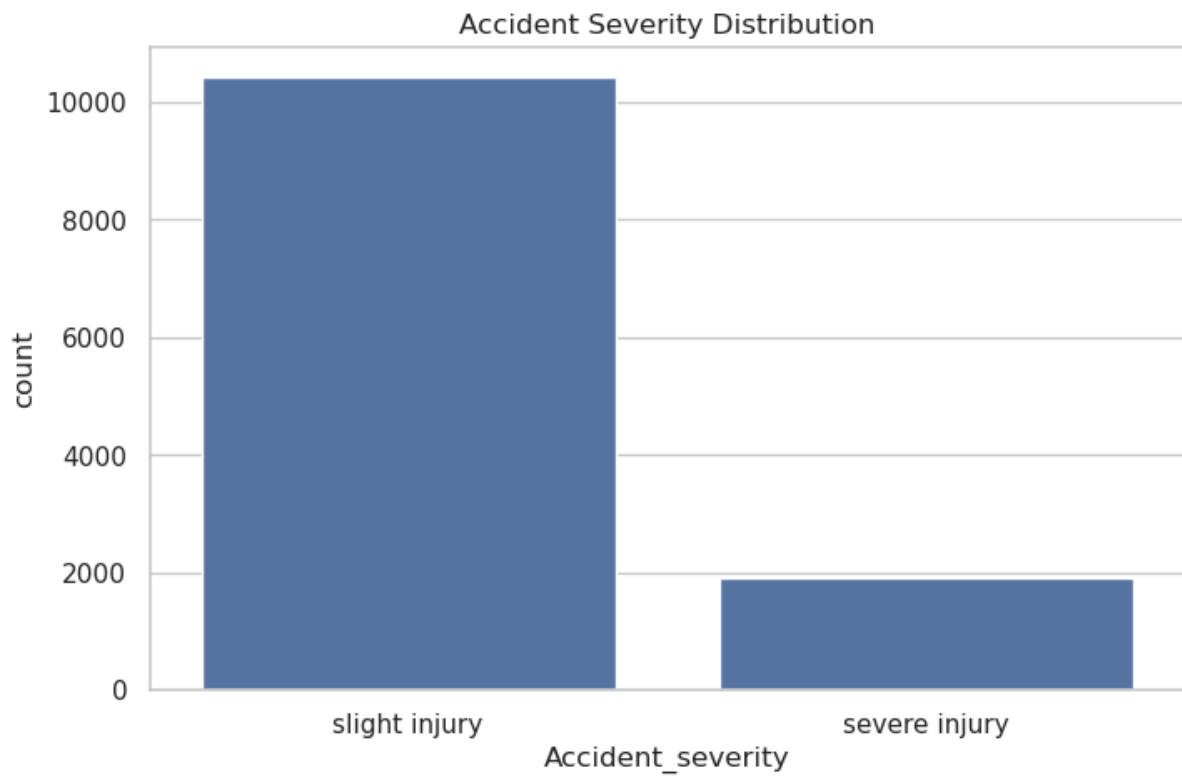
$$F1_{score} = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

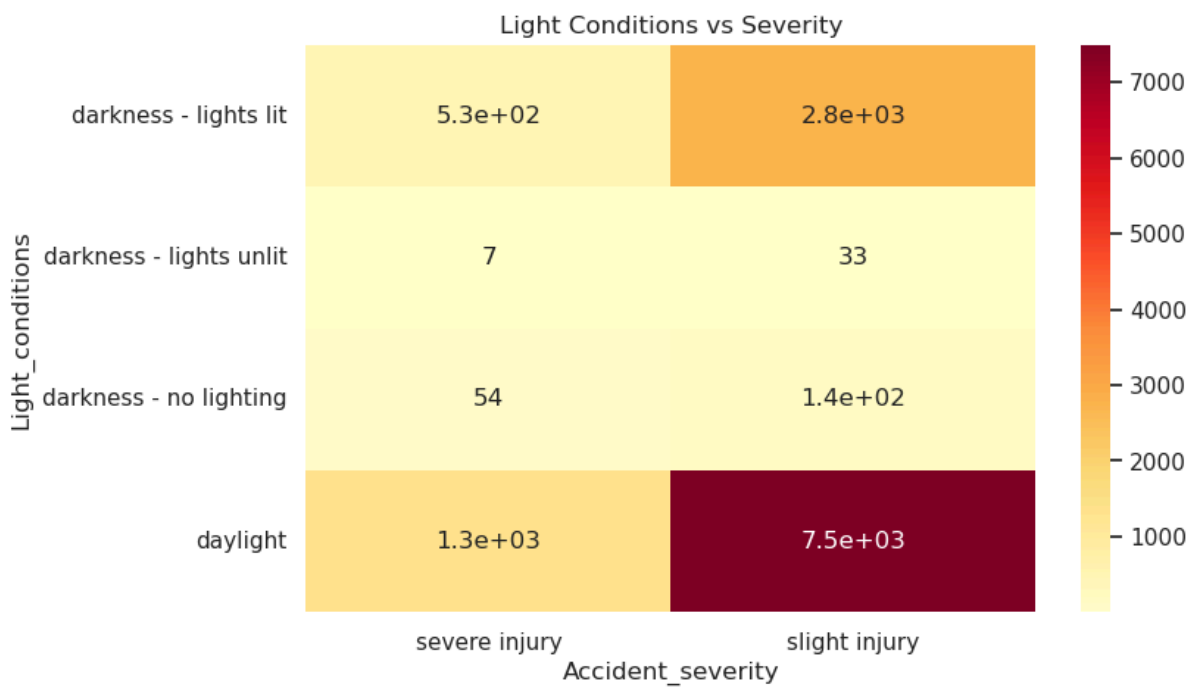
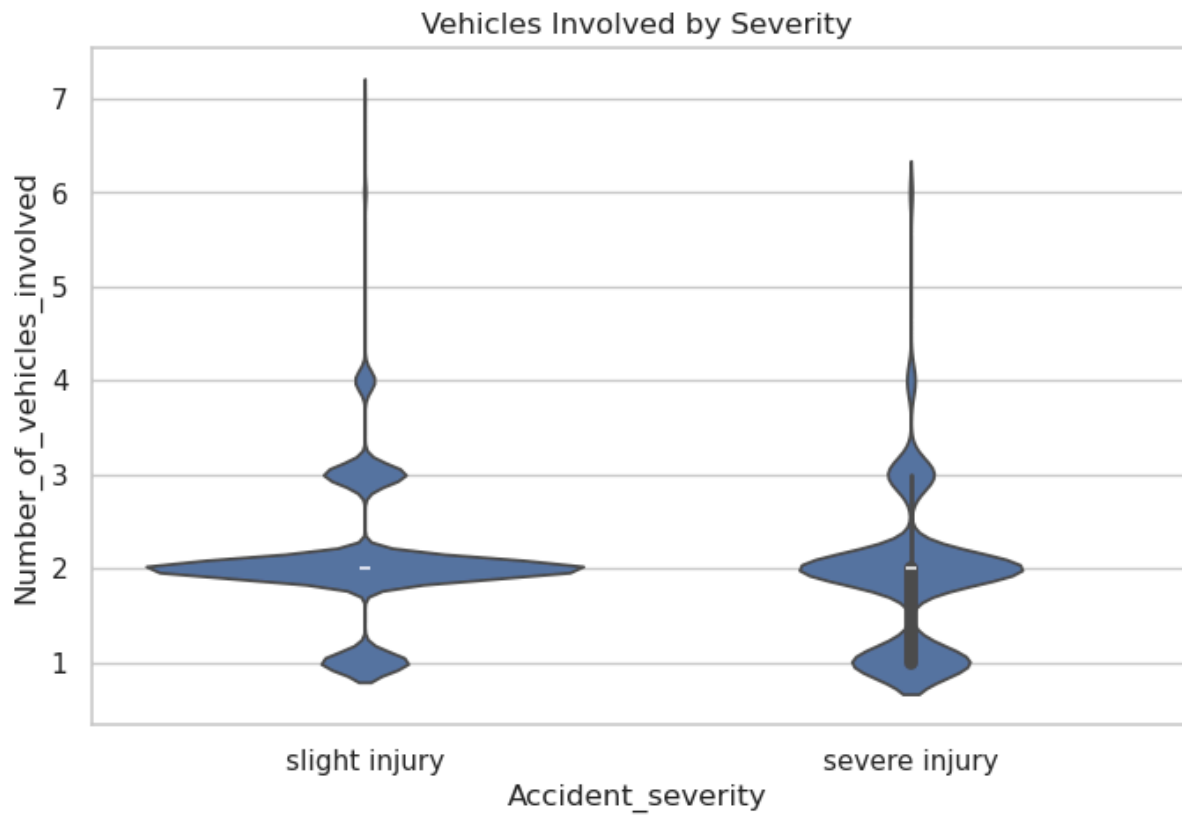
5. Confusion Matrix

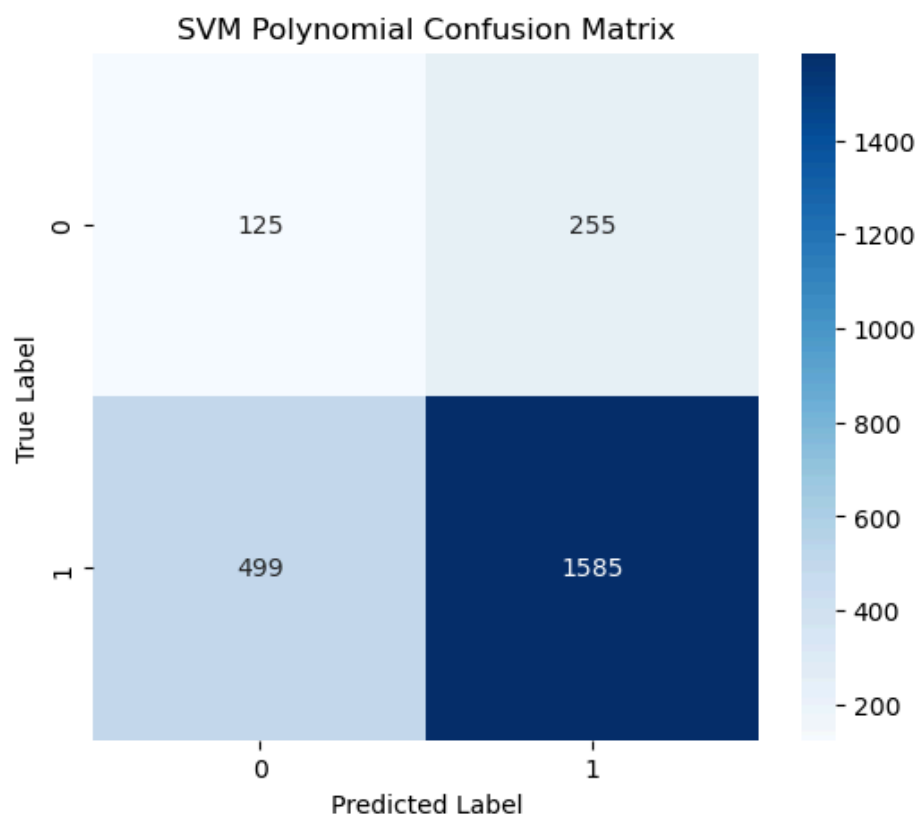
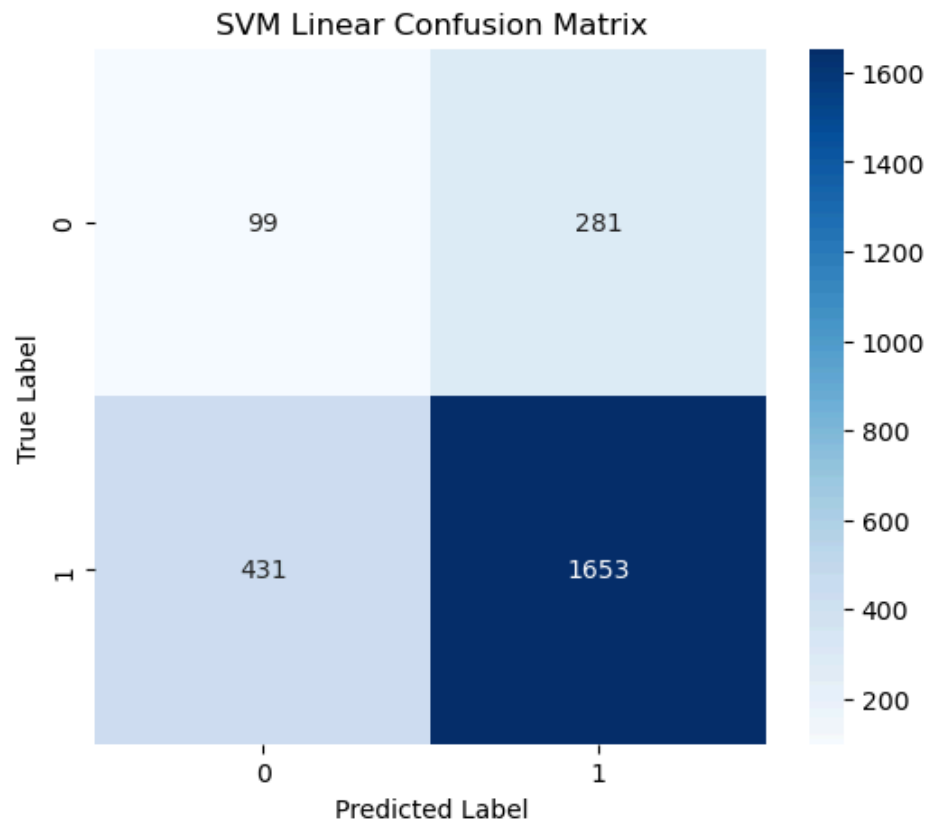
A Confusion Matrix is used to evaluate classification performance by showing the number of correct and incorrect predictions made by the model for each class. It helps analyze false positives and false negatives in accident severity prediction.

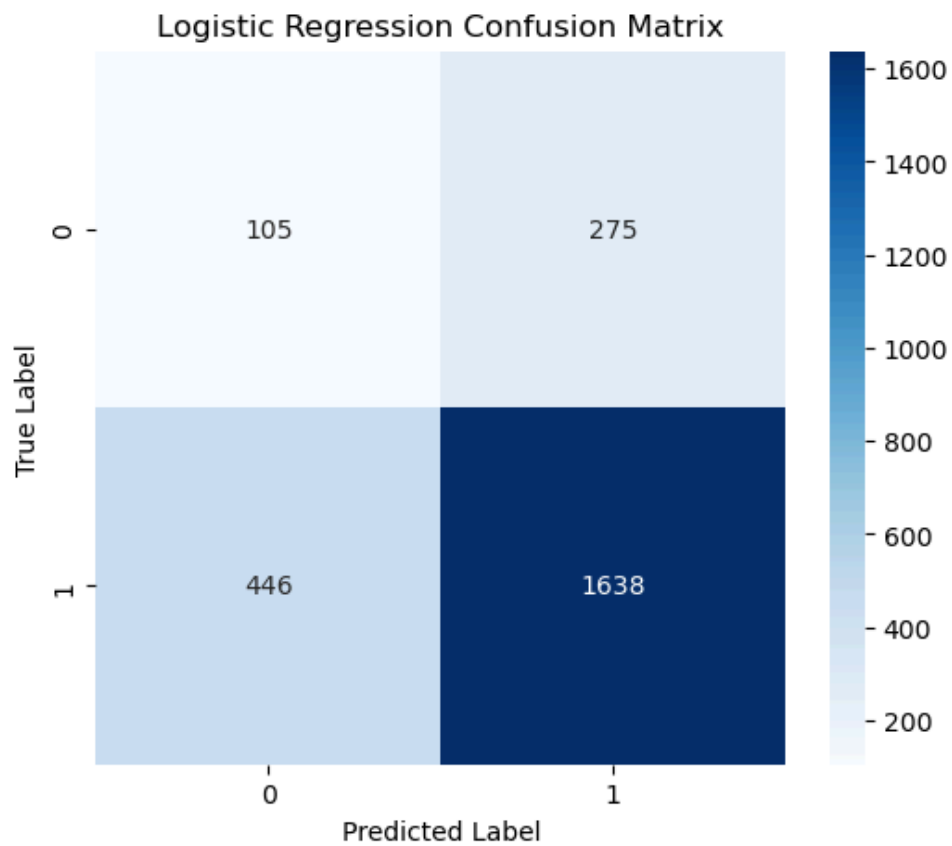
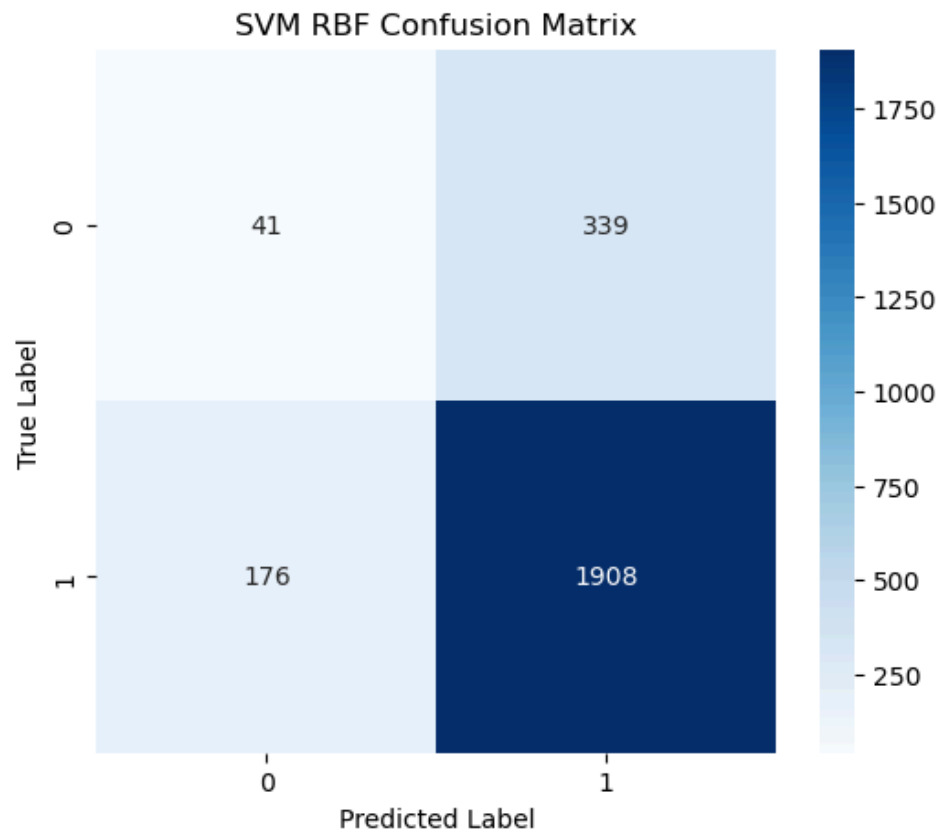
Preliminary Results and Visualizations

- The dataset showed a strong class imbalance where slight injury cases were significantly higher than severe injury cases.
- Exploratory Data Analysis (EDA) was performed using bar plots, count plots, and heatmaps to understand feature distributions and relationships between accident-related factors.
- Data preprocessing techniques such as category reduction, one-hot encoding, feature selection, scaling, and SMOTE balancing improved the quality of training data.
- Multiple machine learning models including Support Vector Machine with Linear, RBF, Polynomial, and Sigmoid kernels and Logistic Regression were trained and evaluated.
- The RBF kernel achieved the highest overall accuracy, while the Polynomial kernel provided better severe injury detection and achieved the best macro-average F1-score.
- Confusion matrices and classification reports were used to compare model performance and analyze prediction behavior for slight and severe injury cases.









Conclusion

The Road Accident Severity Prediction System was successfully developed using machine learning techniques to predict whether an accident would result in slight injury or severe injury. The project included data cleaning, category reduction, exploratory data analysis, preprocessing, feature selection, SMOTE balancing, model training, and implementation.

Different machine learning models including Support Vector Machine with Linear, RBF, Polynomial, and Sigmoid kernels and Logistic Regression were trained and evaluated. Among all models, the Polynomial Kernel SVM performed best in detecting severe accident cases and achieved the best balance between accuracy and minority-class prediction performance. The implementation results also showed that the model successfully classified realistic severe and slight accident scenarios.

The project demonstrates how machine learning can assist in accident severity prediction and support road safety analysis.

Future Scope

- Use larger and real-time accident datasets for improved prediction accuracy.
- Integrate GPS, traffic, and live weather data for real-time accident severity prediction.
- Develop a web or mobile application for practical deployment.
- Apply deep learning techniques for more advanced prediction models.
- Extend the system to predict accident causes and recommend preventive measures.

References

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